APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

KTU CST414 — DEEP LEARNING

COMPLETE 36-LECTURE HANDWRITTEN SLIDES HANDOUT

Printable 2-Slides-Per-Page Edition with Student Derivation Ruling Lines

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LECTURE #01 [Slide 1/2]: Introduction to Neural Networks – Single Layer Perceptrons

Module 1 · Topic 1.1 · CO: CO1 (Understand / Apply)

· Introduction: Artificial Neural Networks (ANNs) are computational models inspired by the human biological nervous system. The fundamental computational unit is the artificial neuron, first conceptualized as the McCulloch-Pitts neuron ...

· Mathematical Model: A Single Layer Perceptron receives an n-dimensional input vector x = [x1, x2, ..., xn]^T and computes the net input z (pre-activation) as the inner product with weight vector w = [w1, w2, ..., wn]^T plus bias b: z = sum...

Key Terms: Artificial Neuron, Threshold Logic Unit (TLU), Perceptron Learning Rule, Linear Separability

Lecture Notes & Derivation Annotations:

LECTURE #01 [Slide 2/2]: Formulation & Industry Case

Mathematical Formulation · Numerical Solution · Production Application

Manual Output Calculation of Single Neuron: Given a 3-dimensional input vector x = [2, 2, 1]^T connected to a single neuron with weight vector w = [0.3, -0.4, 0.5]^T and bias b = -0.1. Calculate the net i...

Result: Net input z = 0.2, Step output = 1, Sigmoid output = 0.5500

Case (Smart Manufacturing & Automated Quality Control): Industrial Defect Screening via Linear Separator — Enables ultra-low-latency (sub-millisecond) hardware defect filtering on microcontrollers without requiring deep GPU har...

Lecture Notes & Derivation Annotations:

LECTURE #02 [Slide 1/2]: Multi-Layer Perceptrons (MLPs) and Representation Power of MLPs

Module 1 · Topic 1.2 · CO: CO1 (Understand / Analyze)

· Introduction: A Multi-Layer Perceptron (MLP) is a class of feedforward artificial neural network consisting of at least three layers of nodes: an input layer, one or more hidden layers, and an output layer. Except for input nodes, eac...
· Architecture And Forward Pass: For an MLP with L layers: Layer 0: Input vector a^(0) = x For layer l = 1, 2, ..., L: z^(l) = W^(l) * a^(l-1) + b^(l) a^(l) = sigma(z^(l)) where W^(l) is an (n_l x n_{l-1}) weight matrix, b^(l) is an (n_l x 1) bias v...

Key Terms: Input Layer, Hidden Layer, Output Layer, Universal Approximation Theorem

Lecture Notes & Derivation Annotations:

LECTURE #02 [Slide 2/2]: Formulation & Industry Case

Mathematical Formulation · Numerical Solution · Production Application

Trainable Parameter Count in Multi-Layer Perceptron: An MLP has an input layer with 784 features, a first hidden layer with 256 neurons, a second hidden layer with 128 neurons, and an output layer with 10 classes....

Result: Total Trainable Parameters = 235,146

Case (Fintech & Credit Risk Assessment): Retail Loan Default Prediction Engine — Detects complex non-linear repayment risk signals that linear logistic regression models miss, cutting default rates by ...

Lecture Notes & Derivation Annotations:

LECTURE #03 [Slide 1/2]: Activation Functions, Risk Minimization and Loss Functions

Module 1 · Topic 1.3 · CO: CO1 (Understand / Apply)

· Activation Functions: 1. Sigmoid (Logistic): sigma(z) = 1 / (1 + e^(-z)), Range: (0, 1) Derivative: sigma'(z) = sigma(z) * (1 - sigma(z)), Max derivative = 0.25 at z = 0. Drawbacks: Vanishing gradient for |z| >> 0; Not zero-centered ...

· Risk Minimization: Statistical Learning Theory defines expected risk R(theta) as the expected loss over the true data distribution p_data(x, y): R(theta) = E_{(x,y)~p_data} [L(f(x; theta), y)] Since p_data is unknown, Empirical Risk Min...

Key Terms: Activation Function, Sigmoid, Hyperbolic Tangent (Tanh), Rectified Linear Unit (ReLU)

Lecture Notes & Derivation Annotations:

LECTURE #03 [Slide 2/2]: Formulation & Industry Case

Mathematical Formulation · Numerical Solution · Production Application

Softmax Probability and Cross-Entropy Loss Calculation: A deep neural network outputs logits z = [2.0, 1.0, -1.0]^T for a 3-class classification problem. The true class label is Class 1 (one-hot vector y = [1, 0, 0]^...

Result: Softmax probabilities = [0.7054, 0.2595, 0.0351]^T; Loss L = 0.3490

Case (Computer Vision & Edge Automotive Systems): Autonomous Vehicle Traffic Sign Recognition — Softmax probabilities allow the automotive decision control unit to establish confidence thresholds (e.g. only brake if ...

Lecture Notes & Derivation Annotations:

LECTURE #04 [Slide 1/2]: Training MLPs with Backpropagation

Module 1 · Topic 1.4 · CO: CO1 (Understand / Apply)

· Introduction: Backpropagation (Rumelhart, Hinton, Williams, 1986) is the foundational training algorithm for multi-layer neural networks. It efficiently computes the partial derivative of the scalar loss function L with respect to eve...

· Mathematical Derivation: Consider layer l with activation a^(l) = sigma(z^(l)), where z^(l) = W^(l) * a^(l-1) + b^(l). Define the Error Sensitivity (Delta) vector for layer l: delta^(l) = del L / del z^(l) 1. Fundamental Equation 1 (Output Lay...

Key Terms: Backpropagation, Chain Rule, Error Sensitivity (Delta), Gradient Vector

Lecture Notes & Derivation Annotations:

LECTURE #04 [Slide 2/2]: Formulation & Industry Case

Mathematical Formulation · Numerical Solution · Production Application

Output Layer Error Delta Calculation: An output neuron with Sigmoid activation has pre-activation z = 1.5, giving output a = 0.8176. The ground truth target is y = 1.0. If the loss function is Squar...

Result: dL/da = -0.1824, da/dz = 0.1491, delta = -0.0272

Case (Digital Signal Processing & Real-time Acoustics): Acoustic Noise Cancellation in Aviation Communication — Enables crystal-clear air traffic controller communication in high-decibel turbofan cockpit environments....

Lecture Notes & Derivation Annotations:

LECTURE #05 [Slide 1/2]: Illustration of the Backpropagation Algorithm

Module 1 · Topic 1.5 · CO: CO1 (Apply / Evaluate)

· Network Specification: Standard 2-2-1 Fully Connected MLP Architecture: Input vector: x = [x1, x2]^T = [0.05, 0.10]^T Target output: y = 0.01 (or target vector for 2 outputs) Initial weights from Input to Hidden layer: V = [[v11, v12], [v21,...

· Step By Step Execution Phases: Phase 1: Forward Pass 1. Hidden Neuron 1: z_h1 = (0.15 * 0.05) + (0.20 * 0.10) + 0.35 = 0.0075 + 0.020 + 0.35 = 0.3775 a_h1 = sigma(0.3775) = 1 / (1 + e^(-0.3775)) = 0.59327 2. Hidden Neuron 2: z_h2 = (0.25 * 0....

Key Terms: End-to-End Walkthrough, Forward Evaluation, Backward Sensitivity, Gradient Update

Lecture Notes & Derivation Annotations:

LECTURE #05 [Slide 2/2]: Formulation & Industry Case

Mathematical Formulation · Numerical Solution · Production Application

Single Step Weight Update with Learning Rate: An output layer weight w has current value 0.45. The computed gradient of the loss function with respect to w is dL/dw = 0.08266. If the learning rate is eta = ...

Result: delta_w = -0.04133, w_new = 0.40867

Case (Electric Vehicles & Battery Management Systems (BMS)): Real-Time Battery State-of-Charge (SoC) Calibration in EVs — Maintains accurate driving range estimation over 10-year battery degradation curves without requiring cloud data transfe...

Lecture Notes & Derivation Annotations:

LECTURE #06 [Slide 1/2]: Practical Issues in Training: Overfitting, Vanishing and Exploding Gradients

Module 1 · Topic 1.6 · CO: CO1 (Understand / Analyze)

· Overfitting: Overfitting occurs when a neural network memorizes training data noise and idiosyncratic artifacts rather than learning generalizable underlying patterns. Key Indicators: - Training Loss continues decreasing monotonical...
· Vanishing And Exploding Gradients: Consider a deep network with L layers. By the chain rule, the gradient of the loss with respect to activation at early layer 1 is: del L / del a^(1) = (del L / del a^(L)) * product_{k=1}^{L-1} (del a^(k+1) / del a^(k)) ...

Key Terms: Overfitting, Generalization Gap, Vanishing Gradients, Exploding Gradients

Lecture Notes & Derivation Annotations:

LECTURE #06 [Slide 2/2]: Formulation & Industry Case

Mathematical Formulation · Numerical Solution · Production Application

Gradient Clipping by Norm: During backpropagation of a deep network, the computed gradient vector across all parameters is g = [3.0, -4.0, 12.0]^T. The maximum permissible gradient clippi...
Result: ||g|| = 13.0, Clipping required. Clipped gradient = [1.1538, -1.5385, 4.6154]^T (Norm = 5.0)

Case (Healthcare Informatics & Predictive Medicine): Clinical Mortality Prediction in Intensive Care Units (ICU) — Maintains reliable 24-hour mortality risk scores without overfitting to sparse hospital training records....

Lecture Notes & Derivation Annotations:

LECTURE #07 [Slide 1/2]: Convergence Difficulties, Local and Spurious Optima, Computational Challenges

Module 1 · Topic 1.7 · CO: CO1 (Understand / Analyze)

· Loss Surfaces In Deep Learning: Unlike linear models whose convex loss functions guarantee a unique global minimum, deep neural networks have highly non-convex loss landscapes in millions of dimensions. The landscape is characterized by: 1. Symmetries ...

· Ill Conditioned Hessian: The Hessian matrix H contains second-order partial derivatives: H_ij = del^2 L / (del w_i del w_j). The condition number kappa(H) = lambda_max / lambda_min measures curvature disparity: - If kappa(H) >> 1, the loss surf...

Key Terms: Non-Convex Optimization, Saddle Point, Spurious Local Minima, Hessian Matrix

Lecture Notes & Derivation Annotations:

LECTURE #07 [Slide 2/2]: Formulation & Industry Case

Mathematical Formulation · Numerical Solution · Production Application

Hessian Matrix Condition Number and Curvature Evaluation: At a critical point in weight space, the Hessian matrix of a neural network loss function is computed as H = [[200.0, 0.0], [0.0, 2.0]]. (a) Identify the eigenv...

Result: Eigenvalues = 200.0 and 2.0. Condition number kappa = 100.0 (Ill-conditioned).

Case (Geophysics & Energy Exploration): Seismic Oil & Gas Subsurface Reservoir Inversion — Prevents million-dollar exploratory dry-well drilling by resolving deep geological formation boundaries....

Lecture Notes & Derivation Annotations:

LECTURE #08 [Slide 1/2]: Introduction to Deep Learning and Deep Feed-Forward Networks

Module 2 · Topic 2.1 · CO: CO2 (Understand)

· Introduction: Deep learning is a subfield of machine learning based on artificial neural networks with representation learning. The adjective 'deep' refers to the use of multiple layers in the network. A deep feedforward network (DFN)...
· Depth Vs Width: Theoretical justification for depth: 1. Exponential Efficiency: Certain functions requiring an exponential number of hidden units in a single-layer network can be approximated to arbitrary accuracy using a deep network w...

Key Terms: Deep Learning, Deep Feedforward Network (DFN), Hierarchical Feature Extraction, Circuit Complexity

Lecture Notes & Derivation Annotations:

LECTURE #08 [Slide 2/2]: Formulation & Industry Case

Mathematical Formulation · Numerical Solution · Production Application

Layer Dimensions and Memory Footprint in Deep Feed-Forward Network: A deep feedforward network has architecture [100, 50, 25, 10, 2]. Each parameter and activation is stored as a 32-bit floating-point number (4 bytes). Calculate...
Result: Weights = 6,520, Biases = 87, Total Params = 6,607 (25.81 KB), Batch Activation Memory = 21.75 KB

Case (Smart Industry 4.0 & Industrial Internet of Things (IIoT)): Industrial High-Dimensional Predictive Maintenance — Enables proactive component replacement prior to catastrophic gearbox seizure, saving over $250,000 per turbine incident...

Lecture Notes & Derivation Annotations:

LECTURE #09 [Slide 1/2]: Training Deep Models: Optimization and Regularization Overview

Module 2 · Topic 2.2 · CO: CO2 (Understand)

· Optimization Vs Machine Learning: In traditional numerical optimization, the objective is to minimize a known function J(theta). In machine learning, our true goal is to minimize the expected risk over the unseen data-generating distribution p_data(x, y)...
· Regularization Definition: Ian Goodfellow defines regularization as: 'Any modification we make to a learning algorithm that is intended to reduce its generalization error but not its training error.' Regularization Taxonomy in Deep Learning: 1. P...

Key Terms: Optimization vs Generalization, Surrogate Loss Function, Regularization, Generalization Error

Lecture Notes & Derivation Annotations:

LECTURE #09 [Slide 2/2]: Formulation & Industry Case

Mathematical Formulation · Numerical Solution · Production Application

Empirical Risk vs Surrogate Loss Evaluation: A binary classifier evaluates 4 test samples with true labels y in {0, 1}. The predicted probabilities p = P(y=1) are [0.90, 0.40, 0.20, 0.85] for true labels [...
Result: 0-1 Classification Error = 0.50 (50%), Cross-Entropy Surrogate Loss = 0.7855

Case (Quantitative Finance & Algorithmic Execution): Real-Time Stock Market Algorithmic Trading — Prevents catastrophic portfolio drawdown during unexpected market volatility spikes....

Lecture Notes & Derivation Annotations:

LECTURE #10 [Slide 1/2]: Gradient Descent (GD) and GD with Momentum

Module 2 · Topic 2.3 · CO: CO2 (Understand / Apply)

· Gradient Descent: Standard Gradient Descent (Cauchy, 1847) updates parameters theta in the direction of the negative gradient: theta_{t+1} = theta_t - eta * grad_theta J(theta_t) Failure Modes: 1. Ravines (Ill-conditioned curvature): The...

· Momentum Algorithm: Polyak (1964) introduced the heavy-ball / momentum method, simulating a physical particle with mass moving down a potential energy surface subject to friction. Mathematical Formulation: v_{t+1} = beta * v_t + (1 - beta)...

Key Terms: Batch Gradient Descent, Momentum, Velocity Vector, Friction Coefficient (Beta)

Lecture Notes & Derivation Annotations:

LECTURE #10 [Slide 2/2]: Formulation & Industry Case

Mathematical Formulation · Numerical Solution · Production Application

Two-Step Momentum Velocity and Weight Calculation: An optimizer with momentum coefficient beta = 0.9 and learning rate eta = 0.1 starts with initial parameter w_0 = 5.0 and initial velocity v_0 = 0.0. The gradie...

Result: At step 1: v_1 = -0.20, w_1 = 4.80. At step 2: v_2 = -0.36, w_2 = 4.44.

Case (Robotics & Real-Time Motion Control): Robotic Arm Inverse Kinematics Trajectory Planning — Reduces robotic pick-and-place cycle times on automotive assembly lines by 24%....

Lecture Notes & Derivation Annotations:

LECTURE #11 [Slide 1/2]: Nesterov Accelerated Gradient (NAG) and Stochastic Gradient Descent (SGD)

Module 2 · Topic 2.4 · CO: CO2 (Understand / Apply)

· Nesterov Accelerated Gradient: Yurii Nesterov (1983) modified classical momentum by computing the gradient not at the current position theta_t, but at the 'look-ahead' position (theta_t + beta * v_t): NAG Equations: 1. Look-ahead projection: theta_lo...

· Stochastic Gradient Descent: Standard Batch GD computes the exact gradient over all N training examples: cost O(N) per step. For millions of samples, this is computationally prohibitive. Stochastic Gradient Descent (Robbins & Monro, 1951) approxima...

Key Terms: Nesterov Accelerated Gradient, Look-Ahead Step, Stochastic Gradient Descent, Mini-Batch SGD

Lecture Notes & Derivation Annotations:

LECTURE #11 [Slide 2/2]: Formulation & Industry Case

Mathematical Formulation · Numerical Solution · Production Application

Nesterov Look-Ahead Position and Gradient Evaluation: An optimizer using Nesterov momentum has current parameter w_t = 2.0, current velocity v_t = -0.5, and momentum factor beta = 0.9. The loss function is J(w) = 0...

Result: w_ahead = 1.55, Gradient = 1.55, v_{t+1} = -0.605, w_{t+1} = 1.395

Case (Edge AI & Drone Surveillance Systems): Real-Time Embedded Video Object Tracking — Enables uninterrupted drone camera lock-on for search-and-rescue target tracking in mountain terrain....

Lecture Notes & Derivation Annotations:

LECTURE #12 [Slide 1/2]: Adaptive Learning Rate Methods: AdaGrad, RMSProp, and Adam

Module 2 · Topic 2.5 · CO: CO2 (Understand / Apply)

· Adagrad: AdaGrad (Duchi, Hazan, Singer, 2011) adapts the learning rate to each individual parameter based on the history of past squared gradients: 1. Accumulate squared gradients: s_t = s_{t-1} + g_t odot g_t 2. Parameter update... · Rmsprop: RMSProp (Geoff Hinton, Coursera Lecture 6e, 2012) fixes AdaGrad's premature freezing by replacing the monotonic sum with an Exponentially Weighted Moving Average (EWMA): 1. Update discounted second moment: s_t = beta_2 *...

Key Terms: AdaGrad, RMSProp, Adam (Adaptive Moment Estimation), Second Moment Accumulator

Lecture Notes & Derivation Annotations:

LECTURE #12 [Slide 2/2]: Formulation & Industry Case

Mathematical Formulation · Numerical Solution · Production Application

AdaGrad vs RMSProp Two-Step Accumulator Comparison: A parameter w has initial value 1.0. At step 1 and step 2, the observed gradients are g1 = 4.0 and g2 = 4.0. The base learning rate is eta = 0.5 and epsilon = 1...
Result: AdaGrad: s_2 = 32.0, eta_eff = 0.0884. RMSProp: s_2 = 3.04, eta_eff = 0.2868.

Case (Autonomous Robotics & Edge Spatial Computing): Autonomous Drone LiDAR Simultaneous Localization and Mapping (SLAM) — Enables rapid convergence of 3D spatial pose graph estimation in sub-second real-time constraints....

Lecture Notes & Derivation Annotations:

LECTURE #13 [Slide 1/2]: Regularization Techniques: L1, L2, Early Stopping, and Dataset Augmentation

Module 2 · Topic 2.6 · CO: CO2 (Understand / Apply)

· L1 And L2 Regularization: Regularized Objective Function: J_tilde(w) = J(w) + alpha * Omega(w) 1. L2 Regularization (Weight Decay / Ridge): Omega(w) = (1/2) * ||w||_2^2 = (1/2) * sum_i w_i^2 Gradient: grad J_tilde = grad J + alpha * w U...

· Early Stopping: Early Stopping is the most widely used regularization technique in deep learning. Protocol: 1. At the end of each epoch, evaluate loss on a hold-out validation set. 2. If validation loss improves, save model parameters t...

Key Terms: L1 Regularization (Lasso), L2 Regularization (Weight Decay / Ridge), Sparsity, Weight Shrinkage

Lecture Notes & Derivation Annotations:

LECTURE #13 [Slide 2/2]: Formulation & Industry Case

Mathematical Formulation · Numerical Solution · Production Application

L2 Weight Decay Shrinkage Step Calculation: A neural network weight currently has value w_t = 0.80. The unregularized loss gradient with respect to this weight is dJ/dw = 0.10. The learning rate is eta = ...

Result: Shrinkage factor = 0.990, Decayed weight = 0.7920, Final weight w_{t+1} = 0.7870

Case (Healthcare Diagnostics & Radiomics): Medical Imaging CT Scan Lung Nodule Detection — Prevents catastrophic overfitting to patient scanner artifacts, boosting lung cancer early detection sensitivity to 96.4...

Lecture Notes & Derivation Annotations:

LECTURE #14 [Slide 1/2]: Parameter Sharing, Parameter Tying, and Injecting Noise at Input

Module 2 · Topic 2.7 · CO: CO2 (Understand / Apply)

· Parameter Sharing And Tying: 1. Parameter Sharing: Instead of assigning unique independent weights to every connection, parameter sharing forces multiple components to share the exact same parameter value. - Primary Example: Convolutional Neural Net...

· Injecting Noise At Input: Adding random noise epsilon ~ N(0, sigma^2 * I) to the input vector x during training: x_tilde = x + epsilon Mathematical Analysis (Bishop, 1995): For a linear or smooth model f(x) trained with squared error loss, minim...

Key Terms: Parameter Sharing, Parameter Tying, Tied Weights, Input Noise Injection

Lecture Notes & Derivation Annotations:

LECTURE #14 [Slide 2/2]: Formulation & Industry Case

Mathematical Formulation · Numerical Solution · Production Application

Parameter Reduction Calculation in Tied Autoencoder: An autoencoder maps an input vector of dimension n_in = 784 into a compressed latent bottleneck of dimension n_h = 64, and then reconstructs the 784-dimensional...

Result: Untied = 101,200 params. Tied = 51,024 params (49.58% parameter reduction).

Case (Audio Signal Processing & In-Cabin Infotainment): Noise-Resistant Automatic Speech Recognition in High-Speed Rail — Maintains 94% command recognition accuracy without acoustic degradation or word drops under 300 km/h train speeds....

Lecture Notes & Derivation Annotations:

LECTURE #15 [Slide 1/2]: Ensemble Methods, Model Averaging, Bagging and Boosting in Deep Learning

Module 2 · Topic 2.8 · CO: CO2 (Understand / Apply)

· Faculty Syllabus Flag: FACULTY NOTICE: In the approved KTU CST414 teaching plan (Page 12 of DEEP LEARNING.pdf), topics 2.7 and 2.8 carry the identical title: 'Parameter sharing and tying, Injecting noise at input, Ensemble methods'. To ensure ...
· Model Averaging Mathematics: Ensemble learning combines the predictions of k distinct models {f_1(x), f_2(x), ..., f_k(x)}: f_ens(x) = (1 / k) * sum_{i=1}^k f_i(x) Variance Reduction Theorem: Suppose each model f_i has error epsilon_i with variance...

Key Terms: Ensemble Learning, Model Averaging, Bootstrap Aggregation (Bagging), Boosting

Lecture Notes & Derivation Annotations:

LECTURE #15 [Slide 2/2]: Formulation & Industry Case

Mathematical Formulation · Numerical Solution · Production Application

Theoretical Ensemble Error Variance Calculation: An ensemble is composed of k = 5 independently trained deep neural networks. Each individual network has an expected squared error variance of v = 0.20. The ave...
Result: Ensemble Error = 0.080 (60% error reduction compared to individual model error 0.20)

Case (Competitive Machine Learning & Financial Risk Modeling): Kaggle Grandmaster Competition Winning Pipeline — Ensemble consistently outscores any individual model by 0.015-0.030 AUC, making the decisive difference between 1st plac...

Lecture Notes & Derivation Annotations:

LECTURE #16 [Slide 1/2]: Dropout and Parameter Initialization

Module 2 · Topic 2.9 · CO: CO2 (Understand / Apply)

· Dropout Formulation: Dropout (Srivastava et al., JMLR 2014) is a computationally efficient regularization technique that prevents feature co-adaptation. 1. Training Phase: For each training example and each training step, each hidden unit i...

· Parameter Initialization: Why not initialize weights to zero? Symmetry problem: all neurons compute identical forward activations and receive identical backpropagated gradients. Why not large random weights? Activations explode or saturate (vanis...

Key Terms: Dropout, Bernoulli Mask, Inverted Dropout, Co-Adaptation

Lecture Notes & Derivation Annotations:

LECTURE #16 [Slide 2/2]: Formulation & Industry Case

Mathematical Formulation · Numerical Solution · Production Application

Inverted Dropout Expected Value Verification: A hidden layer produces activation vector h = [2.0, 4.0, 6.0, 8.0]^T. During training, Inverted Dropout with retention probability p = 0.5 is applied. The gener...

Result: Classical: [2.0, 0.0, 6.0, 0.0]^T (requires test scaling). Inverted: [4.0, 0.0, 12.0, 0.0]^T (no test scaling required).

Case (Computational Biology & Structural Bioinformatics): Protein Folding & Biomolecular Structure Prediction — Enables drug discovery scientists to flag uncertain protein loop predictions before committing to wet-lab synthesized as...

Lecture Notes & Derivation Annotations:

LECTURE #17 [Slide 1/2]: Convolutional Neural Networks: The Convolution Operation

Module 3 · Topic 3.1 · CO: CO3 (Understand / Apply)

· Convolution Operation: Continuous 1D Convolution: s(t) = (x * w)(t) = integral_{-infinity}^{infinity} x(a) * w(t - a) da Discrete 2D Mathematical Convolution (with Kernel Flip): S(i, j) = (I * K)(i, j) = sum_m sum_n I(i - m, j - n) * K(m, n) ...

· Feature Map And Filters: A filter kernel is a small matrix (e.g. 3x3, 5x5) whose elements act as feature detectors: - Horizontal Sobel Filter: K = [[-1, -2, -1], [0, 0, 0], [1, 2, 1]] - Vertical Sobel Filter: K = [[-1, 0, 1], [-2, 0, 2], [-1, ...

Key Terms: Convolution, Cross-Correlation, Filter Kernel, Feature Map

Lecture Notes & Derivation Annotations:

LECTURE #17 [Slide 2/2]: Formulation & Industry Case

Mathematical Formulation · Numerical Solution · Production Application

Manual 2D Discrete Convolution Calculation: Given a 4x4 input image I = [[1, 2, 0, 1], [0, 1, 3, 2], [2, 0, 1, 0], [1, 1, 0, 2]] and a 3x3 kernel K = [[1, 0, -1], [1, 0, -1], [1, 0, -1]]. Calculate the va...

Result: Feature Map S = [[-1, 0], [-1, -2]]

Case (Semiconductor Manufacturing): Silicon Wafer Defect Inspection — Prevents defective silicon chips from reaching automotive and aerospace electronic engine control units....

Lecture Notes & Derivation Annotations:

LECTURE #18 [Slide 1/2]: Motivation for CNNs and Pooling Operations

Module 3 · Topic 3.2 · CO: CO3 (Understand / Apply)

· Three Motivations For Cnns: 1. Sparse Interactions: Kernel size k << input size m. Receptive fields are local, reducing storage from O(m*n) to O(k*n) and slashing FLOPs. 2. Parameter Sharing: Reusing kernel weights across all positions slashes para...

· Pooling Operations: Pooling summarizes nearby outputs: - Max Pooling: S(i, j) = max_{(p,q) in R} I(i+p, j+q) (captures dominant activations). - Average Pooling: S(i, j) = (1/|R|) sum_{(p,q) in R} I(i+p, j+q) (smooth background summary). Poo...

Key Terms: Sparse Interactions, Parameter Sharing, Equivariance to Translation, Max Pooling

Lecture Notes & Derivation Annotations:

LECTURE #18 [Slide 2/2]: Formulation & Industry Case

Mathematical Formulation · Numerical Solution · Production Application

Max and Average Pooling Calculation: Given 4x4 matrix M = [[12, 20, 30, 0], [8, 16, 2, 14], [4, 6, 8, 10], [18, 22, 12, 14]], perform 2x2 pooling with stride 2: (a) Compute Max Pooling, and (b) Com...

Result: Max Pool = [[20, 30], [22, 14]], Avg Pool = [[14.0, 11.5], [12.5, 11.0]]

Case (Biometrics & Airport Security): Biometric Facial Recognition Gates — Enables seamless passenger boarding without requiring rigid motionless posture alignment....

Lecture Notes & Derivation Annotations:

LECTURE #19 [Slide 1/2]: Convolution and Pooling as an Infinitely Strong Prior

Module 3 · Topic 3.3 · CO: CO3 (Understand / Analyze)

· Bayesian Prior Perspective: In Bayesian statistics, a prior probability distribution p(theta) reflects our beliefs about which parameter configurations are plausible before observing any training data. A prior is 'weak' or 'soft' if it assigns pos...

· Convolution And Pooling As Infinitely Strong Prior: A Convolutional layer can be viewed as a fully connected layer with an infinitely strong prior specifying that: 1. All weights connecting non-spatially adjacent pixels must be identically ZERO: p(W_ij != 0) = 0 for |i - ...

Key Terms: Bayesian Prior, Infinitely Strong Prior, Inductive Bias, Parameter Constraint

Lecture Notes & Derivation Annotations:

LECTURE #19 [Slide 2/2]: Formulation & Industry Case

Mathematical Formulation · Numerical Solution · Production Application

Weight Matrix Density under Infinitely Strong Prior: Consider a 1D input of length N = 100 mapped to an output of length N = 100. (a) In a dense MLP, calculate the total number of weight connections in the 100x100...

Result: Non-zero entries = 500; Zero-constrained entries = 9,500 (95.0% forced to zero).

Case (Bioinformatics & Computational Genomics): Genomic Variant Pathogenicity Modeling — Applying the correct inductive bias prevents misinterpreting unstructured patient records while capturing sequential gen...

Lecture Notes & Derivation Annotations:

LECTURE #20 [Slide 1/2]: Variants of the Convolution Function

Module 3 · Topic 3.4 · CO: CO3 (Understand / Apply)

· Strided Convolution: Standard convolution slides the kernel by 1 pixel at each step. A strided convolution slides the kernel by s pixels (s > 1): S(i, j) = sum_m sum_n I(i*s + m, j*s + n) * K(m, n) Effect: Downsamples the feature map by a fa...

· Zero Padding Types: For input dimension W, kernel size K, stride s, and zero-padding p: Output Dimension: W_out = floor((W - K + 2*p) / s) + 1 1. Valid Padding (p = 0): No padding added. W_out = (W - K + 1). Feature maps shrink at eve...

Key Terms: Strided Convolution, Valid Padding, Same Padding, Full Padding

Lecture Notes & Derivation Annotations:

LECTURE #20 [Slide 2/2]: Formulation & Industry Case

Mathematical Formulation · Numerical Solution · Production Application

Convolution Output Shape Calculation across Padding Modes: An input feature map has spatial resolution W_in = 32x32. A convolutional layer uses filter size K = 5x5 with stride s = 1. Calculate the output spatial dimensi...

Result: Valid = 28x28, Same = 32x32, Full = 36x36

Case (Autonomous Vehicles & Scene Parsing): DeepLab Autonomous Driving Semantic Segmentation — Enables pixel-accurate sidewalk, curb, and pedestrian segmentation without spatial detail loss....

Lecture Notes & Derivation Annotations:

LECTURE #21 [Slide 1/2]: Structured Outputs and Diverse Data Types in CNNs

Module 3 · Topic 3.5 · CO: CO3 (Understand / Apply)

· Data Type Taxonomy: CNNs process multi-dimensional grid data across diverse domains: 1. 1D Sequential Grids (Channels x Time): - Audio waveforms (1D amplitude sampled at 16 kHz or 44.1 kHz). - Biomedical ECG / EEG sensor telemetry. ...

· Structured Outputs: Rather than outputting a single categorical scalar y in {1, ..., K}, structured prediction tasks output high-dimensional structured arrays: 1. Semantic Segmentation (Dense Pixel Classification): - Output is a 2D labe...

Key Terms: Structured Outputs, Multi-Channel Tensors, 1D Convolution, 3D Volumetric Convolution

Lecture Notes & Derivation Annotations:

LECTURE #21 [Slide 2/2]: Formulation & Industry Case

Mathematical Formulation · Numerical Solution · Production Application

3D Volumetric Convolution Output Dimension and Parameter Count: A medical MRI brain scan has volumetric resolution 128x128x64 voxels (Height x Width x Depth) with 1 input channel. A 3D convolutional layer applies 32 filters ...

Result: Output dimensions = 126 x 126 x 62 (32 channels); Total parameters = 896

Case (Medical Radiology & Neurosurgical Navigation): Brain Glioblastoma Tumor Segmentation in 3D MRI — Enables neurosurgeons to achieve millimeter-accurate tumor resection while avoiding critical motor cortex brain tissue....

Lecture Notes & Derivation Annotations:

LECTURE #22 [Slide 1/2]: Efficient Convolution Algorithms

Module 3 · Topic 3.6 · CO: CO3 (Understand / Apply)

· Computational Bottleneck: Standard 2D Convolution on an input of size D_F x D_F with M channels, producing N channels with kernel size D_K x D_K requires: Total Multiplications = D_F * D_F * M * N * D_K * D_K For a 56x56 feature map with M=256, N...

· Depthwise Separable Convolutions: MobileNet (Howard et al., 2017) factorizes standard convolution into two independent steps: 1. Depthwise Convolution: Applies a single spatial filter per input channel (M filters of size D_K x D_K x 1). Cost = D_F * D...

Key Terms: Efficient Convolutions, Depthwise Separable Convolution, Pointwise Convolution, Fast Fourier Transform (FFT)

Lecture Notes & Derivation Annotations:

LECTURE #22 [Slide 2/2]: Formulation & Industry Case

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MobileNet Depthwise Separable FLOP Calculation: A convolutional layer operates on an input feature map of size 28x28 with M = 64 input channels and produces N = 128 output channels using 3x3 filters (stride =...

Result: Standard = 57.8M ops. Separable = 6.87M ops (88.11% computation reduction, ~8.4x speedup).

Case (Consumer Electronics & Edge Computer Vision): Smart Home Edge Camera Face Unlock — Enables instant biometric door unlock without draining battery or sending private home video streams to the cloud....

Lecture Notes & Derivation Annotations:

LECTURE #23 [Slide 1/2]: Recurrent Neural Networks: Computational Graphs and RNN Design

Module 4 · Topic 4.1 · CO: CO4 (Understand / Apply)

· Recurrent Equations And Unfolding: A Recurrent Neural Network processes sequential data x^(1), x^(2), ..., x^(tau) by maintaining an internal hidden state memory vector h^(t) that summarizes past history. State Update Equation: h^(t) = tanh(W_hh * h^(t-...

· Backpropagation Through Time: Total sequence loss is the sum of losses at each time step: L = sum_{t=1}^tau L^(t). By the multivariable chain rule, the gradient of loss at step tau with respect to hidden state at early step t is a product of Jacobia...

Key Terms: Recurrent Neural Network (RNN), Hidden State, Computational Graph Unfolding, Backpropagation Through Time (BPTT)

Lecture Notes & Derivation Annotations:

LECTURE #23 [Slide 2/2]: Formulation & Industry Case

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Vanilla RNN Hidden State and Output Calculation: A simple RNN has parameters W_xh = [0.5, 0.2], W_hh = 0.8, b_h = 0.1, W_hy = 1.2, and b_y = -0.3. The activation function for the hidden state is Tanh, and the ...

Result: Step 1: h_1 = 0.7616, y_1 = 0.6139. Step 2: h_2 = 0.6408, y_2 = 0.4690.

Case (Renewable Energy & Power Distribution): Smart Power Grid Wind Energy Output Forecasting — Enables power grid operators to balance battery storage and spin up backup generation without blackouts....

Lecture Notes & Derivation Annotations:

LECTURE #24 [Slide 1/2]: Encoder-Decoder Sequence-to-Sequence Architectures

Module 4 · Topic 4.2 · CO: CO4 (Understand / Apply)

· Seq2Seq Architecture: Standard RNNs require input length T_x and output length T_y to be strictly equal or synchronized. Many tasks require mapping a sequence of length T_x to an output sequence of different length T_y (e.g. Machine Translati...

· Information Bottleneck And Teacher Forcing: 1. Information Bottleneck Problem: Compressing an entire source sentence of arbitrary length (e.g. 50 words) into a single fixed-dimensional vector c (e.g. 512 floats) creates an extreme bottleneck. Model performance det...

Key Terms: Sequence-to-Sequence (Seq2Seq), Encoder RNN, Decoder RNN, Context Vector

Lecture Notes & Derivation Annotations:

LECTURE #24 [Slide 2/2]: Formulation & Industry Case

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Context Vector Compression Ratio in Seq2Seq: An English sentence of 40 words is processed by an Encoder RNN. Each word is represented by a 300-dimensional word embedding vector. The Encoder condenses the s...

Result: Input = 48,000 bytes. Context = 2,048 bytes. Compression ratio = 23.44x.

Case (Medical Informatics & Cross-Lingual Healthcare): Automated Clinical Discharge Summary Translation — Eliminates language barriers in rural hospital discharge, cutting medication dosing misunderstandings by 35%....

Lecture Notes & Derivation Annotations:

LECTURE #25 [Slide 1/2]: Deep Recurrent Networks and Recursive Neural Networks

Module 4 · Topic 4.3 · CO: CO4 (Understand / Apply)

· Deep Stacked Rnns: A standard RNN is deep in time (across unrolled sequence steps), but shallow in spatial representation (only one hidden layer per time step). A Deep (Stacked) RNN stacks multiple recurrent layers vertically: - Layer 1 r...

· Bidirectional Rnns: Schuster & Paliwal (1997) introduced Bidirectional RNNs (BiRNNs) to resolve the limitation that standard causal RNNs only know past context. Equations: 1. Forward RNN: h_fwd^(t) = f(W_xf * x^(t) + W_ff * h_fwd^(t-1)) ...

Key Terms: Deep Recurrent Network, Stacked RNN, Bidirectional RNN (BiRNN), Forward State

Lecture Notes & Derivation Annotations:

LECTURE #25 [Slide 2/2]: Formulation & Industry Case

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Trainable Parameter Count in a 3-Layer Deep Stacked RNN: A deep stacked RNN has an input dimension of x_dim = 128. It contains 3 stacked recurrent hidden layers, each having hidden dimension h_dim = 256. The top hidde...

Result: Layer 1 = 98,560; Layer 2 = 131,328; Layer 3 = 131,328; Output = 2,570. Total = 363,786 parameters.

Case (Clinical NLP & Biomedical Information Extraction): Named Entity Recognition (NER) in Electronic Health Records (EHR) — Extracts structured clinical adverse drug event records from unstructured physician notes with 93.8% F1-score....

Lecture Notes & Derivation Annotations:

LECTURE #26 [Slide 1/2]: Modern RNNs: Long Short-Term Memory (LSTM)

Module 4 · Topic 4.4 · CO: CO4 (Understand / Apply)

· Lstm Architecture And Equations: Hochreiter & Schmidhuber (1997) designed the LSTM to maintain a linear memory cell state c^(t) where error gradients can flow backward across time without exponential decay. Given input x^(t) and previous hidden state h...

· Mathematical Resolution Of Vanishing Gradients: In a vanilla RNN, del h^(t) / del h^(t-1) = W_hh^T * diag(1 - h^2). Repeated multiplication of matrices with eigenvalues < 1 causes exponential decay. In an LSTM, the gradient flowing along the cell state is: del c^(t) ...

Key Terms: Long Short-Term Memory (LSTM), Cell State, Constant Error Carousel (CEC), Forget Gate

Lecture Notes & Derivation Annotations:

LECTURE #26 [Slide 2/2]: Formulation & Industry Case

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Manual Step-by-Step LSTM Gate and State Calculation: An LSTM has scalar dimensions (x_dim = 1, h_dim = 1). At step t = 1, previous hidden state is h_0 = 0.5 and previous cell state is c_0 = 2.0. The input is x_1 =...

Result: f_1 = 0.6682, i_1 = 0.5866, c_tilde_1 = 0.6911, Cell State c_1 = 1.7418, Hidden State h_1 = 0.6387

Case (Critical Care Medicine & Early Warning Clinical Systems): Long-Term ICU Patient Sepsis Onset Prediction — Enables preemptive broad-spectrum antibiotic intervention, reducing ICU sepsis mortality by 28%....

Lecture Notes & Derivation Annotations:

LECTURE #27 [Slide 1/2]: Gated Recurrent Unit (GRU) and Modern Sequence Architectures

Module 4 · Topic 4.5 · CO: CO4 (Understand / Apply)

· Gru Architecture And Equations: Kyunghyun Cho et al. (2014) introduced the Gated Recurrent Unit (GRU) as a streamlined, computationally efficient alternative to LSTM. Key Structural Differences from LSTM: 1. No separate Cell State: GRU merges cell sta...

· Lstm Vs Gru Comprehensive Comparison: Comparison Dimension | LSTM | GRU Gates | 3 (Forget, Input, Output) | 2 (Reset, Update) Internal States | 2 (Cell c + Hidden h) | 1 (Merged Hidden h) Parameters per Unit | 4 * (d_x + d_h + 1) * d_h | 3 * (d_x + d_h + 1) ...

Key Terms: Gated Recurrent Unit (GRU), Reset Gate, Update Gate, Merged Hidden State

Lecture Notes & Derivation Annotations:

LECTURE #27 [Slide 2/2]: Formulation & Industry Case

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Manual Step-by-Step GRU State Calculation: A GRU cell has scalar dimensions (x_dim = 1, h_dim = 1). At step t = 1, the previous hidden state is h_0 = 0.6. The input is x_1 = 0.5. Parameters are: Reset ga...

Result: Reset gate r_1 = 0.5987, Update gate z_1 = 0.5915, Candidate h_tilde_1 = 0.5281, Final Hidden State h_1 = 0.5575

Case (Smart Home IoT & Ultra-Low-Power Edge Audio): On-Device Voice Assistant Wake-Word Detection — Maintains 98.7% wake-word detection accuracy while drawing less than 15 milli-Watts on battery power....

Lecture Notes & Derivation Annotations:

LECTURE #28 [Slide 1/2]: Deep Learning Applications: Computer Vision

Module 5 · Topic 5.1 · CO: CO5 (Understand / Apply)

· Computer Vision Taxonomy: Deep learning has transformed computer vision across four primary task tiers: 1. Image Classification: Predict a single discrete category label for the entire image (e.g. ResNet on ImageNet 1k). 2. Object Localization: P...

· Object Detection Architectures: 1. Two-Stage Detectors (Faster R-CNN, Ren et al., 2015): - Stage 1: Region Proposal Network (RPN) slides across convolutional feature maps to generate candidate object proposals with anchor boxes. - Stage 2: RoI Pooling/...

Key Terms: Computer Vision, Object Detection, Two-Stage Detectors, Single-Stage Detectors (YOLO)

Lecture Notes & Derivation Annotations:

LECTURE #28 [Slide 2/2]: Formulation & Industry Case

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Manual Intersection over Union (IoU) Calculation: Two bounding boxes are defined on an image grid: Ground Truth Box A: [x1 = 20, y1 = 20, x2 = 80, y2 = 80] Predicted Box B: [x1 = 40, y1 = 40, x2 = 100, y2 = ...

Result: Area_A = 3600, Area_B = 3600, Inter = 1600, Union = 5600, IoU = 0.2857 (28.57%)

Case (Automotive Safety & Self-Driving Vehicles): Autonomous Highway Collision Avoidance & Emergency Braking — Slashes forward vehicle collision accidents by 73% in poor visibility rain and fog conditions....

Lecture Notes & Derivation Annotations:

LECTURE #29 [Slide 1/2]: Deep Learning Applications: Speech Recognition

Module 5 · Topic 5.2 · CO: CO5 (Understand / Apply)

· Speech Representation: Speech is an analog acoustic pressure wave sampled at high frequencies (16 kHz = 16,000 samples/sec). Acoustic Feature Extraction: 1. Short-Time Fourier Transform (STFT): Apply a Hamming window of 25ms with 10ms hop siz...

· Connectionist Temporal Classification: Challenge: Audio frames T (e.g. 1,000 frames) far exceed target character length U (e.g. 50 characters), and exact frame-level alignments are unknown. CTC Solution (Graves et al., 2006): 1. Introduces a special 'Blank' ...

Key Terms: Automatic Speech Recognition (ASR), Mel Frequency Cepstral Coefficients (MFCC), Mel Filterbank, Connectionist Temporal Classification (CTC)

Lecture Notes & Derivation Annotations:

LECTURE #29 [Slide 2/2]: Formulation & Industry Case

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Word Error Rate (WER) Calculation: An Automatic Speech Recognition (ASR) system transcribes a recorded meeting sentence. The reference transcript contains N = 10 words: Reference: 'the weather i...

Result: S = 0, D = 2, I = 1. Total Errors = 3. Word Error Rate (WER) = 30.0%.

Case (Public Safety & Telecommunications): Real-Time Emergency Dispatch Voice Call Transcription — Enables emergency first responders to dispatch ambulances 45 seconds faster, saving critical lives....

Lecture Notes & Derivation Annotations:

LECTURE #30 [Slide 1/2]: Deep Learning Applications: Natural Language Processing

Module 5 · Topic 5.3 · CO: CO5 (Understand / Apply)

· Nlp Task Taxonomy: Deep learning models natural language across diverse granularities: 1. Document / Sentence-Level Classification: - Sentiment Analysis: Input text -> Output sentiment label (Positive, Neutral, Negative). - Topic Cat...
· Perplexity Metric: Perplexity (PPL) is the standard quantitative metric for evaluating language models. It is defined as the exponentiated average negative log-likelihood per token: PPL = exp(- (1 / T) * sum_{t=1}^T ln P(w_t | w_{<t})) =...

Key Terms: Natural Language Processing (NLP), Autoregressive Language Modeling, Perplexity (PPL), Sentiment Classification

Lecture Notes & Derivation Annotations:

LECTURE #30 [Slide 2/2]: Formulation & Industry Case

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Perplexity Calculation from Cross-Entropy Loss: A neural language model evaluates a test document containing T = 1,000 words. The total negative log-likelihood of the document under natural logarithm (base e)...
Result: Average Loss = 3.50 nats, Perplexity PPL = 33.12. Competing Model Loss = 3.00 nats.

Case (Fintech & Quantitative Regulatory Analysis): Automated Financial SEC 10-K Filing Risk Assessment — Enables asset managers to detect emerging corporate insolvency risks weeks before quarterly earnings announcements....

Lecture Notes & Derivation Annotations:

LECTURE #31 [Slide 1/2]: Common Word Embedding: CBOW and Word2Vec

Module 5 · Topic 5.4 · CO: COS (Understand / Apply)

· One Hot Vs Distributed Embeddings: One-Hot Encoding: - Represents each word as a sparse binary vector of vocabulary size |V| (e.g. 100,000 dimensions). - Fatal Flaw: All word vectors are orthogonal: v_cat^T * v_dog = 0. No notion of semantic distance or s...

· Word2Vec Cbow Vs Skipgram: Tomas Mikolov et al. (Google, 2013) introduced Word2Vec with two complementary architectures: 1. Continuous Bag-of-Words (CBOW): - Objective: Predict a target center word w_t given context words in a window: {w_{t-2}, w...

Key Terms: Word Embeddings, One-Hot Encoding, Word2Vec, Continuous Bag of Words (CBOW)

Lecture Notes & Derivation Annotations:

LECTURE #31 [Slide 2/2]: Formulation & Industry Case

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Cosine Similarity Calculation Between Word Vectors: Two word embedding vectors in 3-dimensional semantic space are given as: v_doctor = [0.8, 0.6, 0.0]^T v_hospital = [0.6, 0.8, 0.0]^T v_banana = [0.0, 0.1, 0.9]^...

Result: cos(doc, hosp) = 0.9600 (Strongly Related). cos(doc, ban) = 0.0663 (Unrelated).

Case (Retail E-Commerce & Product Discovery): E-Commerce Semantic Search & Recommendation Engine — Increases search conversion rates by 19% by overcoming the vocabulary mismatch problem....

Lecture Notes & Derivation Annotations:

LECTURE #32 [Slide 1/2]: Common Word Embedding: Global Vectors for Word Representation (GloVe)

Module 5 · Topic 5.5 · CO: CO5 (Understand / Apply)

· Glove Motivation: Prior to GloVe (2014), representation learning was divided into two distinct paradigms: 1. Global Matrix Factorization Methods (LSA): Leveraged global corpus statistics, but performed poorly on local word analogy tasks. ...

· Mathematical Foundation And Cooccurrence Ratios: Let X_ij be the count of times word j appears in the context of word i in the corpus. Let P_ij = P(j | i) = X_ij / X_i be the probability that word j appears in the context of word i. The core insight of GloVe is that s...

Key Terms: Global Vectors (GloVe), Co-occurrence Matrix, Co-occurrence Probability Ratio, Log-Bilinear Model

Lecture Notes & Derivation Annotations:

LECTURE #32 [Slide 2/2]: Formulation & Industry Case

Mathematical Formulation · Numerical Solution · Production Application

GloVe Weighting Function f(x) Calculation: The GloVe weighting function is defined as f(x) = (x / x_max)^alpha if x < x_max else 1.0, with x_max = 100 and alpha = 0.75. Calculate the weight f(x) assigned...

Result: f(0) = 0.0, f(16) = 0.2530, f(81) = 0.8538, f(5000) = 1.0000

Case (Legal Tech & Corporate Compliance): Legal Contract Discovery & Precedent Retrieval — Slashes paralegal legal discovery search time from 8 hours to 15 minutes per case....

Lecture Notes & Derivation Annotations:

LECTURE #33 [Slide 1/2]: Current Research Areas: Autoencoders and Representation Learning

Module 5 · Topic 5.6 · CO: CO5 (Understand / Apply)

· Autoencoder Architecture: An Autoencoder is a self-supervised neural network trained to reconstruct its own input: 1. Encoder: Maps input x in R^d to a latent code h in R^k: h = f(x) = sigma(W_e * x + b_e) 2. Decoder: Reconstructs input from la...

· Autoencoder Variants: 1. Undercomplete Autoencoders: - Constrains latent dimension k < d (information bottleneck). - Theorem: If the encoder and decoder are linear and loss is MSE, the undercomplete autoencoder learns the exact same subspace ...

Key Terms: Autoencoder, Undercomplete Autoencoder, Latent Bottleneck, Reconstruction Loss

Lecture Notes & Derivation Annotations:

LECTURE #33 [Slide 2/2]: Formulation & Industry Case

Mathematical Formulation · Numerical Solution · Production Application

Kullback-Leibler (KL) Divergence Sparsity Penalty Calculation: In a sparse autoencoder, the target average activation is set to rho = 0.05. A hidden neuron j has computed average activation rho_hat_j = 0.10 across a trainin...

Result: KL(0.05 || 0.10) = 0.01671. Target penalty = 0.0000.

Case (Railway Safety & Predictive Infrastructure): Industrial High-Speed Rail Track Flaw Ultrasonic Anomaly Detection — Detects internal rail fatigue fractures before structural derailment failures occur....

Lecture Notes & Derivation Annotations:

LECTURE #34 [Slide 1/2]: Current Research Areas: Representation Learning

Module 5 · Topic 5.7 · CO: CO5 (Understand / Analyze)

· Manifold Hypothesis: The Manifold Hypothesis states that real-world high-dimensional data (e.g. 1000x1000 pixel images in R^1,000,000) concentrates its probability mass almost entirely on low-dimensional manifolds embedded within the high-di...

· Disentangled Representations: Bengio, Courville, and Vincent (2013) define an ideal representation as a 'Disentangled Representation'. Properties: - Each coordinate axis of the latent representation h corresponds to an independent real-world explana...

Key Terms: Representation Learning, Manifold Hypothesis, Disentangled Representations, Self-Supervised Learning

Lecture Notes & Derivation Annotations:

LECTURE #34 [Slide 2/2]: Formulation & Industry Case

Mathematical Formulation · Numerical Solution · Production Application

InfoNCE Contrastive Loss Calculation: In a self-supervised contrastive learning step, an anchor representation z_a is compared against 1 positive augmented sample z_p and 2 negative samples {z_n1, z...

Result: P_pos = 0.6819, InfoNCE Loss L = 0.3829

Case (Digital Pathology & Oncology Diagnostics): Self-Supervised Medical Pathology Slide Pre-Training — Achieves 95.2% diagnostic accuracy on ultra-rare cancers where obtaining hundreds of labeled slides is medically impossi...

Lecture Notes & Derivation Annotations:

LECTURE #35 [Slide 1/2]: Current Research Areas: Boltzmann Machines and Deep Belief Networks

Module 5 · Topic 5.8 · CO: CO5 (Understand / Analyze)

· Energy Based Models: An Energy-Based Model associates a scalar energy E(x) with each configuration of variables. High-probability states correspond to low energy levels. The Boltzmann Distribution defines probability via the Gibbs / Boltzma...

· General Boltzmann Machine: A Boltzmann Machine (1985) is an undirected graphical model containing visible units v (observed data) and hidden units h (latent representations): Energy Function: E(v, h) = - v^T * W * h - (1/2) * v^T * L * v - (1/2) *...

Key Terms: Energy-Based Model (EBM), Boltzmann Distribution, Partition Function Z, Restricted Boltzmann Machine (RBM)

Lecture Notes & Derivation Annotations:

LECTURE #35 [Slide 2/2]: Formulation & Industry Case

Mathematical Formulation · Numerical Solution · Production Application

RBM Energy and Hidden Activation Probability Calculation: An RBM has 2 visible units and 2 hidden units. Parameters are: Weight matrix W = [[0.8, -0.4], [0.5, 0.6]] Visible bias b = [0.1, -0.2]^T Hidden bias c = [-0.1,...

Result: P(h_1=1|v) = 0.6682, P(h_2=1|v) = 0.4750, Energy E(v, h) = -0.70

Case (Recommender Systems & E-Commerce): Collaborative Filtering Netflix Movie Recommendation — Was a core winning algorithmic component of the famous million-dollar Netflix Prize competition (Salakhutdinov et al., 2...

Lecture Notes & Derivation Annotations:

LECTURE #36 [Slide 1/2]: Current Research Areas: Deep Belief Networks (DBNs) and Course Synthesis

Module 5 · Topic 5.9 · CO: CO5 (Understand / Evaluate)

· Dbn Architecture: A Deep Belief Network (Hinton et al., 2006) is a deep generative hybrid graphical model formed by stacking multiple Restricted Boltzmann Machines (RBMs). Hybrid Graphical Structure: - Top two layers form an UNDIRECTED b...

· Contrastive Divergence: The log-likelihood gradient for an RBM is: del ln P(v) / del W = E_data [v * h^T] - E_model [v * h^T] - Positive Phase (Data-dependent expectation): Easily evaluated using training sample v and P(h|v). - Negative Pha...

Key Terms: Deep Belief Network (DBN), Hybrid Graphical Model, Contrastive Divergence (CD-k), Greedy Layer-Wise Pre-training

Lecture Notes & Derivation Annotations:

LECTURE #36 [Slide 2/2]: Formulation & Industry Case

Mathematical Formulation · Numerical Solution · Production Application

Single Step Contrastive Divergence (CD-1) Weight Update Calculation: An RBM has 2 visible units and 1 hidden unit with scalar weights W = [0.2, -0.1]^T (biases are zero). A training sample is v_0 = [1.0, 1.0]. Learning rate is et...

Result: Positive stats = [0.525, 0.525]^T, Negative stats = [0.271, 0.250]^T, Updated W = [0.3272, 0.0374]^T

Case (Heavy Industry & Vibration Telemetry Analysis): Industrial Turbine Fault Diagnosis using Deep Belief Networks — Enables predictive maintenance that avoids multi-million-dollar turbine rotor catastrophic blade fracture....

Lecture Notes & Derivation Annotations: